

Charles Joshua T. Uy

Tacloban City, Leyte, Philippines

Phone: +63 917 150 4686 | Email: charlesjoshuaay@gmail.com

[Portfolio: cjoy.dev](https://cjoy.dev) | [GitHub: github.cjoy.dev](https://github.com/cjoy.dev) | [LinkedIn: linkedin.cjoy.dev](https://www.linkedin.com/in/cjoy.dev) | [Facebook: facebook.cjoy.dev](https://www.facebook.com/cjoy.dev)

EDUCATION

Ateneo de Manila University

BS Computer Science

- Director's Lister — First Honors
- Fr. Bienvenido F. Nebres, SJ Science Academic Scholar

Loyola Heights, Quezon City

2024 – Present

Philippine Science High School — Eastern Visayas Campus

Core Science (Advanced Physics) | Math Level 2 (Advanced)

- Graduated With High Honors; ranked 2.5 of 88; G12 GWA 1.07/5 (97.68/100)
- Director's Lister for 20 consecutive quarters (Grade 8 through Grade 12)
- PHINMA Academic Excellence Award; Excellence Awards in Mathematics, Computer Science, and Research
- Computer Science Elective (G11) & Engineering Elective (G12); SAT 1480 (Math 780, English 700)

Palo, Leyte

2018 – 2024

INTERNATIONAL HONORS & AWARDS

International Young Researchers Tournament *international research competition*

Gold Award 2024

13th Regional Congress, SEAMEO Search for Young Scientists (SSYS) *SE-Asian research competition*

Finalist 2024

Hong Kong International Computing Olympiad (HKICO), Final Round
international CS olympiad

Gold Medal 2023

International Mathematics Wizard Challenge (IMWiC) *SE-Asian math contest*

Silver Medal 2023

Global Gifted Problem Solving Camp *int'l robotics camp, Gimje, S. Korea*

Leadership Award 2023

International Kangaroo Mathematics Competition

Bronze Award 2021

Hua Luogeng Golden Cup Mathematics Invitational *one of China's toughest contests*

Bronze Award 2019

International Math Open for Young Achievers (IMOYA), Singapore

Silver (Individual & Group) 2017

International Teenagers Mathematics Olympiad (ITMO), National Parallel

Silver Award 2017

VANDA International Science Competition

Bronze Award 2017

International Mathematics Wizard Challenge (IMWiC) *SE-Asian math contest*

Bronze Medal 2016

Australian Mathematics Competition (AMC)

Certificate of Distinction
2015

World Math Invitational, Taiwan

Merit 2014

NATIONAL HONORS & AWARDS

2nd KPMG Academic Innovation Challenge *R.G. Manabat & Co. (KPMG Philippines), with Microsoft & MAP — Team Luzon Visayas Mindanao, 100+ teams*

3rd Place 2026

Start Up QC Squad 3 *Quezon City startup accelerator competition (ongoing)*

Finalist 2026

Blue Hacks 2026 *24-hour hackathon, Ateneo de Manila University*

2nd Runner-Up 2026

HACKFEST 2026: AXIS *10-day hackathon, ADMU Google Developers Group*

2nd Runner-Up 2026

Hacker Cup 2025 2-day hackathon, De La Salle University	2nd Runner-Up 2025
Blue Hacks 2025 24-hour hackathon, Ateneo de Manila University	Top 10 Finisher 2025
Ateneo College Entrance Test (ACET) top 200 of 10,000+ applicants	Director's Lister 2024
7th IMake.WeMake: Create. Innovate. Collaborate national makers competition, DOST-SEI	Finalist (Top 20) 2024
DOST Youth Excellence in Science (YES) Award national STEM achievement award	Awardee 2023
2nd Ateneo Physics Olympiad: Entanglement (APhO) national physics olympiad	Finalist 2023
Philippine Online Physics Olympiad (PhO2) national physics olympiad	Silver Medal 2023
DOST-SEI 6th Indie-Siyensya (Youth Category) national science short-film contest	3rd Place 2023
25th SquEEEze: National EEE HS Competition & Convention electronics & electrical engineering	1st Runner-Up 2023
Laban ng mga Maharlika Science Camp national group science camp	1st Place 2021
Pakaritay regional science quiz bee	1st Place 2021
Metrobank-MTAP-DepEd Math Challenge regional math competition	Top Scorer 2018
DOST Youth Excellence in Science (YES) Award national STEM achievement award	Awardee 2017
DOST Youth Excellence in Science (YES) Award national STEM achievement award	Awardee 2016
National Parallel Challenge for Future Mathematicians	Silver Medal 2016

UNIVERSITY AWARDS

The GUIDON — 2nd Manuel Colayco Awards Outstanding Interactive Award, for A Long Walk Back	Winner 2026
The GUIDON — 2nd Manuel Colayco Awards Rising Guidoneer Award (top first-year staffer)	Finalist 2026
Cyber Flag Rush (CYFR) school-wide CTF (cryptography, digital forensics), CompSAT, 20+ teams — Team Cicada Too	2nd Runner-Up 2025

LEADERSHIP

ARSA Seniors' Send-Off 2026 Secretariat Head. Annual dorm event sending off 100+ graduating seniors.	2026
ARSAFEST SIDLAK 2026 Secretariat Head. Annual dorm event celebrating the diverse people and culture of the Ateneo dorms.	2026
ARSA Year-End Party 2025 Materials & Decorations Head. Official dorm student event with 300+ attendees.	2025
Pisay Math Circle President, Internal Affairs. Led the official PSHS-EVC math club and organized a region-wide math competition.	2023 – 2024
Batch of 88–90 Students (G10, G12) President. Led batch-wide activities, events, and fundraisers.	2021 – 2024

Class of 30 Students (G7, G8, G11)

2018 – 2023

President. Led class activities and yearly community outreaches.

Athletic Council

2021 – 2022

President. Organized the annual school SportsFest for 500+ students.

EXPERIENCE

Surtix

2026 – Present

*Operations Analyst — QA & Automation**Singapore (Remote)*

- Support QA operations for a Singapore-based startup, building automated QA tooling to streamline testing and release workflows

AKIVA Holdings Inc.

2025 – Present

*Freelance Software Developer & Team Lead**Philippines*

- Lead a development team building custom software that automates clients' legacy systems
- Lead developer of Agila, a centralized web app digitizing company-wide requests, forms, and approval workflows

Ateneo Residents Student Association (ARSA)

2025 – Present

*Deputy for Administration & Lead Developer**Ateneo de Manila University*

- Lead the ARSA Digital Ecosystem: a self-hosted hub with a custom URL shortener, event ticketing, admin panels, and an e-commerce shop
- Built an automated GCash OCR receipt-verification system; the shop has processed PHP 250,000+ in total revenue
- Create masterfiles and cross-departmental systems for community events and projects

The Guidon

2025 – Present

*Digital Development Staffer**Ateneo de Manila University*

- Develop web-based interactives for the university's official student publication
- Project lead for A Long Walk Back, a 3D interactive on the 1986 EDSA People Power Revolution — winner, Outstanding Interactive Award, 2nd Manuel Colayco Awards

Computer Society of the Ateneo (CompSat)

2024 – Present

*Developer & Learn2Dev Mentor**Ateneo de Manila University*

- Build web-based projects for clients and student communities
- Mentor and module writer for Learn2Dev: taught and authored modules on modern full-stack web development

Ateneo CODE

2025 – Present

*Member — Consultants for Organization Development & Empowerment**Ateneo de Manila University*

- Provide organizational-development consulting for youth-led organizations to support youth empowerment
- Building CODE Nest, the org's internal digital infrastructure platform

Nissan Tacloban (Palo, Leyte)

2021

*Data Entry Intern**Palo, Leyte*

- One-month internship digitizing and organizing dealership records and customer data for TopCars Autoworld Corporation's Nissan Tacloban branch

PROJECTS

Atelier

2026 – Present

Personal portfolio. An Astro 6 site on Cloudflare Workers combining Svelte, React, Vue, Canvas 2D, and Tailwind into a distinctive ink-on-paper editorial system, centered on a scroll-driven 56×40 generative grid that morphs between identity states — portrait, web, waveform, banig weave, blueprint.

- Ally** 2026 – Present
Developer. AI-powered study companion: parses uploaded syllabi into deadlines, exams, and weak spots, then plans and adapts study blocks through a calendar/task dashboard, a chat assistant, and a focus assistant that blocks distractions and negotiates breaks — with a mobile companion for session alerts.
- Meera** 2026
Developer. AI-powered desktop support assistant pairing a Next.js/React web app with an Electron overlay client for real-time, screen-aware guidance — click-through cursors, arrows, highlights, and chat bubbles rendered over other apps, backed by secure Groq AI routes, validated tool calls, support-ticket workflows, and Drizzle/Cloudflare D1.
- Game of the Generals Online** 2026
Lead Developer. Next.js/Cloudflare web app bringing the Filipino strategy classic Salpakan into a chess-platform-style experience: hidden-piece gameplay, legal move handling, battle resolution, guest play, pass-and-play, solo practice, private room codes, real-time room sync, chat, and move history.
- ARSA Digital Ecosystem** 2025 – Present
Lead Developer. Centralized association hub: custom URL shortener, event + ticketing, admin panels, and an e-commerce shop with automated GCash OCR receipt verification (PHP 250,000+ revenue).
- Harbor PH** 2025 – Present
Lead Developer. Nationwide event platform centralizing discovery and ticketing for diverse gatherings.
- Agila** 2025 – Present
Lead Developer. Digitizes and automates company-wide requests, forms, and approval workflows.
- KonsulTulong** 2025 – Present
Research Lead & Lead Developer. Pre-consultation patient data collection and risk assessment.
- Sim.ula** 2026
Lead Developer — Blue Hacks 2026 (2nd Runner-Up). Urban-policy AI simulation that assesses the impact of proposed Quezon City policies against past outcomes.
- Schrollar** 2026
Lead Developer — Hackfest 2026: Axis (2nd Runner-Up). Explore a topic's research ecosystem as a social feed — papers as posts, citations as replies.
- Navio** 2026
Developer. End-to-end IoT wearable GPS safety tracker: an ESP32-based device captures live GPS, uploads over WiFi with a SIM800L SMS fallback offline, and reports prepaid balance, while a Next.js/TypeScript + Cloudflare D1 web app gives a live Google Maps tracker with device status, speed, satellite count, route history, and upload logs.
- alARMED** 2026
Developer. Physics/IoT project: a WiFi-configurable ESP32 + Arduino alarm clock paired with an autonomous pan-tilt turret with an ESP32-CAM that tracks your face and sprays water to wake you, syncing wirelessly with the alarm body.
- QR/FORGE** 2026
Developer. Ad-free, in-browser QR-code generator and customizer with custom styles, colors, and logo embedding.
- Toyota Valenzuela Community** 2026
Developer. Community platform with events, forums, chat, and photo sharing for local car owners.

Blank Board	2026
Developer. Installable collaborative whiteboard app for chaotic, free-form thinking, with a unique authentication flow.	
ARSAFest Ticketing System	2026
Developer. Online QR-code event ticket generator and verifier.	
Memorize Baybayin	2026
Lead Developer. Anki-style flashcards to master the Baybayin script, powered by a custom spaced-repetition algorithm.	
CODE Nest	2026
Developer. Internal digital infrastructure platform for Ateneo CODE (Consultants for Organization Development & Empowerment).	
PatchPortrait	2026
Developer. Photo-mosaic generator that composes an image from variably-sized tiles instead of a uniform grid.	
Project Sinag	2025
Lead Full-Stack Developer — BlueHacks 2025 (Top 10). Citizen monitoring and reporting with geolocation-tagged images on a map.	
Klutch	2025
Full-Stack & AI Developer — DLSU Hacker Cup 2025 (2nd Runner-Up). Connects users with local blue-collar workers for gigs through a bidding and AI tagging system.	
ARSA FlowerFest Shop	2025
Systems Manager & Developer. Online shop for selling, managing, and fulfilling flower deliveries.	
PDFGetPNG	2025
Developer. Extracts PDF pages as PNG images entirely in the browser.	
Keybound	2024
Lead Developer. Java Swing 2-player arena battler: spell-casting apprentices duel via typed key combos (fire, water, earth, wind) on a tile-based island map, with local/online lobby support, a server-authoritative game loop, custom pixel art, music and sound effects, and a scoreboard.	
DNA Barcode Generator	2024
Lead Developer. Fetches mtDNA from NCBI and renders it visually as a DNA barcode.	
Bio: Tech or Terror	2024
Developer. Educational site on biotechnology in biowarfare, featuring an interactive COVID-scientist simulator.	
ArTest of Fate Online Gallery	2022
Lead Developer. Online gallery for the school-wide SportsFest art competition.	
Automated Receipt Processor & Recorder	2022
Volunteer Developer. Uses OCR to process and record bank, GCash, and Maya receipts for the PSHS-EVC PTA.	

RESEARCH

READSmart Co-author & Lead Developer. AI speech-recognition app for reading assessment and analytics in a rural elementary school.	2026 – Present
Don't Leaf It to Chance Research Leader. Toxicological analysis and DNA barcoding of traditional Filipino products.	2022 – 2024
RAILS: Reservation & Inventory for Laboratory Systems Research Leader & Lead Developer. Full-stack lab-management web app for school laboratory units and similarly scaled systems.	2022 – 2024
Project PAGSACA Research Leader. Vertical farming integrating Plant Microbial Fuel Cells, electroculture, and organic solar cells.	2022 – 2024
TaB Propellers Research Leader. Hybridizing toroidal and bionic propellers for increased efficiency and noise reduction.	2023 – 2024
SPIIn Battery: Spinning Portable Inertial Battery Research Leader. Conceptualization of a portable mechanical battery.	2023
The Good, the Bad, and the Medium Researcher. Case study on student perceptions of teachers.	2022 – 2023

ACTIVITIES & INVOLVEMENT

ARSAound Bass Guitarist. Performed with the dorm music group at gigs from concerts to events.	2026 – Present
ARSAFest Concert Band Bass Guitarist. Live concert band performance.	2026
32nd Int'l Conference on Computers in Education (ICCE) Student Volunteer. Helped facilitate the week-long ICCE 2024.	2024
ECHO Senior Consultant & Mentor. Official CS alternative-learning activity of PSHS-EVC.	2023
Project Teenspired Volunteer & Technical Assistant. National advocacy against teenage pregnancy.	2019 – 2023
Junior Chamber International (JCI) Tacloban Love Member. Community-driven projects.	2022 – 2023
Summer Math Tutorial Math Tutor. Lectured core concepts and prepared daily math activities.	2022
RITS High School Press Futurize Participant. International collaboration to create a culture-showcasing magazine.	2021
Ritsumeikan Global Summit 2020 (Japan) Participant. International summit for cultural exchange.	2020

SELECTED EVENTS & RECOGNITION

OpenAI Build Week Manila

2026

Participant. Received US\$150 in Codex and OpenAI API credits.

CERTIFICATIONS

Machine Learning Specialization (3/3 courses) — DeepLearning.AI & Stanford Online

2024 – 2025

Supervised ML: Regression & Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommenders & RL

Deep Learning Specialization (3/5 courses) — DeepLearning.AI

2025

Neural Networks & Deep Learning; Improving Deep Neural Networks; Structuring ML Projects

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, HTML, CSS

Frameworks: React, Next.js, Vue, Nuxt, SvelteKit, Node.js, Astro, Svelte

Tools & Platforms: Linux, Git, Self-Hosting, Cloudflare (Workers/D1), REST APIs, OCR pipelines

IoT & Electronics: Arduino, ESP32 (incl. ESP32-CAM), sensor/actuator integration, WiFi-connected embedded systems

Focus Areas: Full-Stack Web Development, Backend, Frontend, Machine Learning, AI, Technical Writing

LANGUAGES

English, Filipino, Waray-waray, Bisaya